Optimizing Mordred Descriptors Calculation for Scalable Molecular Feature Extraction

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Introduction

- **Mordred descriptors** [1] work great:
  - molecular fingerprint
  - targets for pretraining neural networks, e.g. CheMeleon [2]
- 1613 physicochemical & topological descriptors
- **problem:** they are too slow, ~0.1s per molecule
- **project goal:** rewrite to be much faster, include in **scikit-fingerprints**

Our improvements

Processing as a fingerprint:

- original Mordred computes each descriptor separately, with very limited reuse of common elements
- we precompute and share components, e.g. atomic properties, subgraphs

Better implementations:

- NumPy vectorized array operations instead of Python loops
- smarter algorithms, e.g. subgraphs enumeration (ESU algorithm), walk counts (adjacency matrix eigenvalues)

Efficient parallelism:

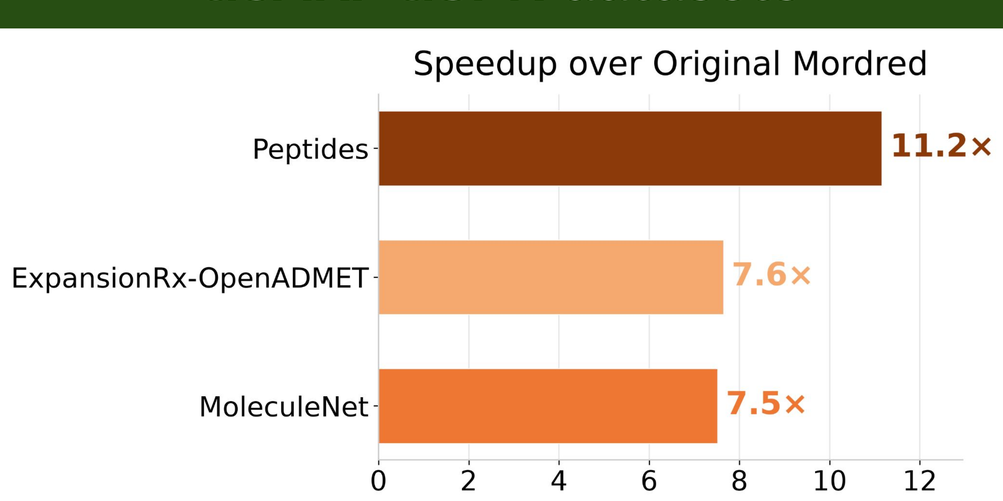
- original Mordred makes a lot of unnecessary inter-process communication
- we have no shared state between molecules and workers, limiting overhead

Individual descriptors speedup

Descriptor family	Features	Original Mordred	Our	Speedup
Extended Topochemical Atom	45	24.75 s	2.65 s	9.3×
Molecular Distance Edge	19	9.75 s	1.35 s	7.2×
Autocorrelation	606	17.11 s	2.61 s	6.6×
EState	316	18.18 s	3.53 s	5.1×
Barysz Matrix	104	19.76 s	4.01 s	4.9×
Chi	56	49.47 s	10.13 s	4.9×
Information Content	42	21.93 s	4.83 s	4.5×
Molecular Id	12	13.39 s	3.43 s	3.9×
Path Count	21	35.36 s	9.42 s	3.8×
Detour Matrix	14	5.98 s	2.03 s	3.0×

- **much faster individual descriptors across the board**
- BACE MoleculeNet dataset, 1513 molecules
- separate descriptors, excludes further gains from shared precomputing

QSAR/QSPR datasets



- **consistent gains, at least 7.5× faster than original**
- mean speedup per benchmark
- varied dataset & molecule sizes, primarily small drug-like

At pretraining scale

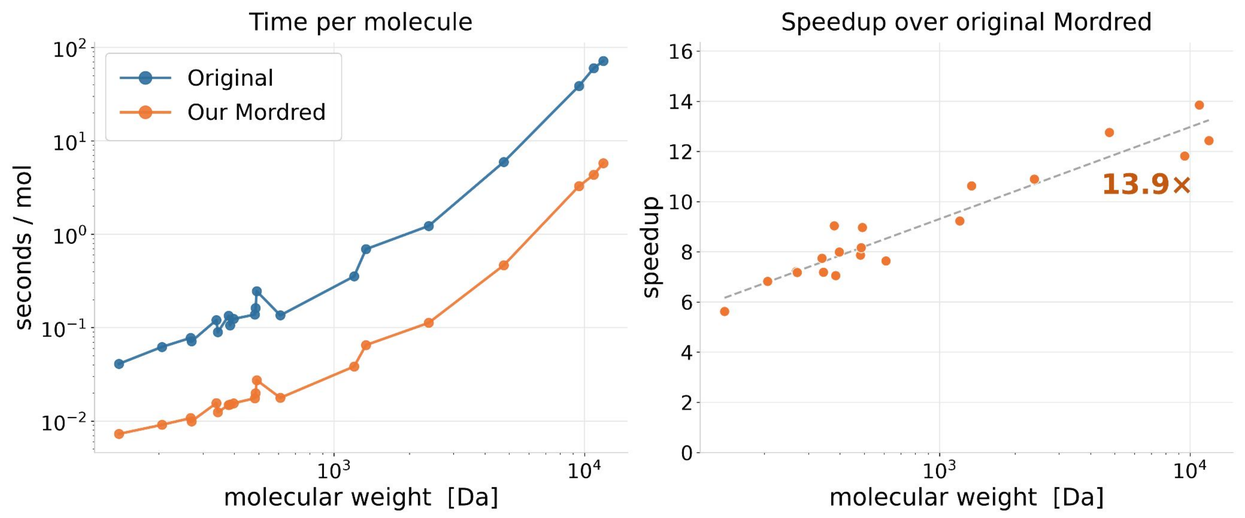
4.5 min
1M mols, 96 workers

3 800 mol/s

16 h
222M mols in MolPILE

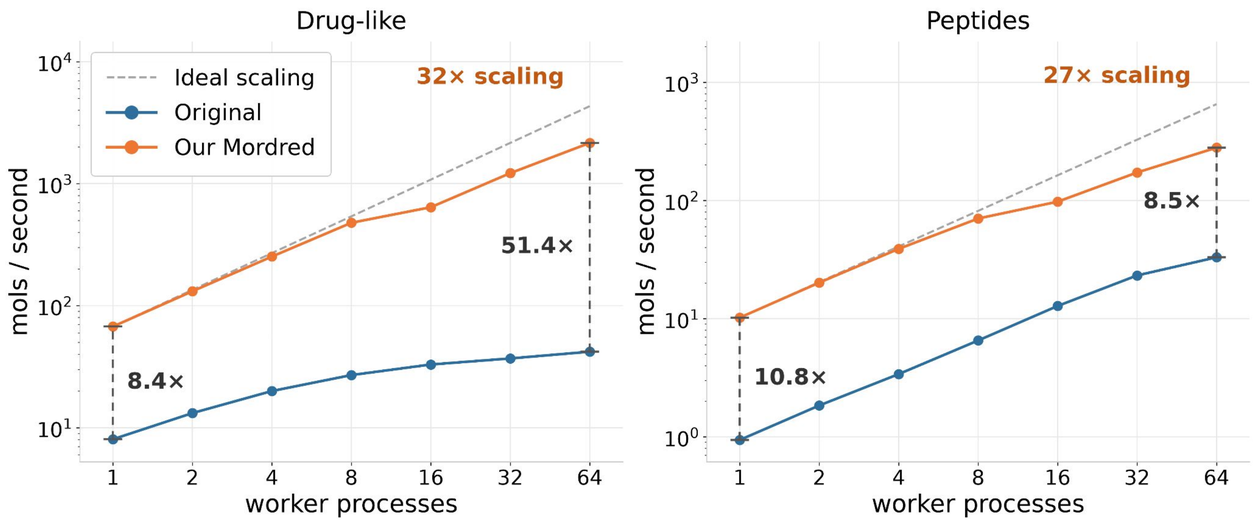
- **our Mordred works well for large-scale computation**
- e.g. enables pretraining CheMeleon on MolPILE [3]

Molecule size scaling



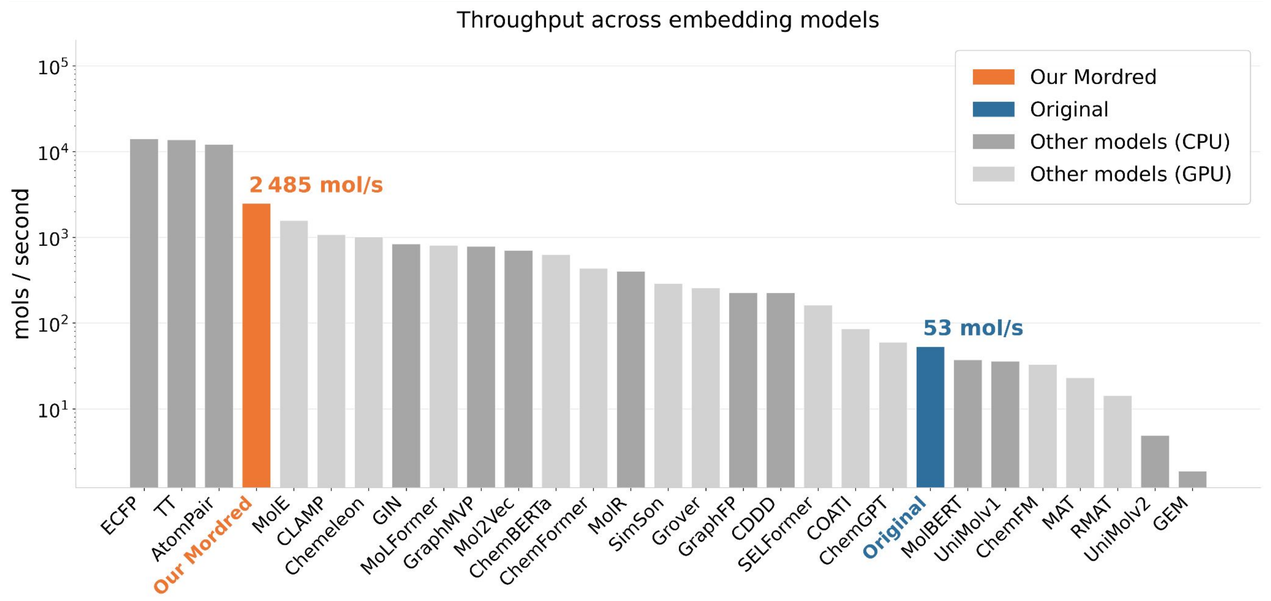
- **very good scaling, bigger gains for larger molecules**
- 5.6× at 140 Da, 13.9× at 11 kDa
- note the log-scales above

Parallel scaling



- **near-linear scaling, on both drug-like molecules and peptides**
- 32× from one to 64 workers on drug-like, 27× on peptides
- HIV dataset (MoleculeNet) and 20-mer peptides

Throughput vs other embeddings



- **better throughput than all neural networks**
- 2 485 mols/s against 53 for the original
- MUV benchmark datasets, 96 CPU workers or RTX 4000 Ada GPU

References

1. Moriwaki, Hirotomo, et al. *"Mordred: a molecular descriptor calculator"*. Journal of Cheminformatics 10.1 (2018): 4.
2. Burns, Jackson W., et al. *"Deep Learning Foundation Models for Low-Data Regimes from Classical Molecular Descriptors"*. Journal of Chemical Information and Modeling.
3. Adamczyk, Jakub, et al. *"MolPILE - large-scale, diverse dataset for molecular representation learning"*. arXiv preprint arXiv:2509.18353 (2025).